￼ Easy Level (10 Questions)

Q1. What does IPsec primarily provide?
A) Caching
B) Encryption and authentication
C) NAT translation
D) DNS resolution
Answer: B

Q2. Which mode of IPsec encrypts the **entire IP packet**?
A) Transport
B) Tunnel
C) Layer 2
D) Full mode
Answer: B

Q3. What type of VPN routes **all traffic** through the tunnel?
A) Half tunnel
B) Split tunnel
C) Full tunnel
D) MPLS
Answer: C

Q4. Which VPN protocol fully supports IPv6?
A) PPTP
B) OpenVPN
C) GRE only
D) Telnet
Answer: B

Q5. Tunnel mode in IPsec adds:
A) A new TCP header
B) A new IP header
C) An Ethernet header
D) DNS header
Answer: B

Q6. What is the **main benefit** of split tunneling?
A) Blocks HTTPS
B) Reduced latency and bandwidth usage
C) Full encryption of all traffic
D) Uses only IPv6
Answer: B

Q7. IPsec operates at which OSI layer?
A) Transport
B) Network
C) Data link
D) Session
Answer: B

Q8. In a Hybrid VPN, combining OpenVPN for users and IPsec for sites gives:
A) Slower VPN
B) No security
C) Flexible architecture
D) Encrypted Ethernet
Answer: C

Q9. Which protocol negotiates IPsec parameters?
A) HTTPS
B) TLS
C) IKE
D) FTP
Answer: C

Q10. In IPv6 VPN, what does **no NAT** result in?
A) More IP conflicts
B) Simplified routing
C) More complex configuration
D) Disconnection
Answer: B

Shall I continue with the **20 medium-level questions**?

You said:

yes

ChatGPT said:

Here are the **20 Medium-Level MCQs** on **Hybrid VPN**, IPsec, IPv6 VPN, and Split/Full Tunnel concepts, with answers and explanations:

� Hybrid VPN – Medium-Level MCQs (20 Questions)

Q11. Which IPsec component ensures **data integrity** and **origin authentication**?
A) ESP
B) AH (Authentication Header)
C) GRE
D) NAT-T
Answer: B
� *AH provides integrity and authentication but not encryption.*

Q12. In a hybrid VPN model, what is typically the reason to **combine GRE with IPsec**?
A) Reduce encryption overhead
B) Support dynamic routing protocols like OSPF or BGP
C) Avoid using public IP addresses
D) Hide IPv4 packets
Answer: B

Q13. Which of the following is **not encrypted** in IPsec's transport mode?
A) TCP header
B) Payload
C) IP header
D) None of the above
Answer: C
Transport mode leaves the original IP header visible.

Q14. In a hybrid VPN setup, what is the advantage of using OpenVPN for remote users?
A) It uses GRE encapsulation
B) It automatically assigns IP addresses
C) It can operate over port 443, bypassing firewalls
D) It does not require encryption
Answer: C

Q15. Which component in IPsec is responsible for **negotiating encryption keys and algorithms**?
A) ESP
B) ISAKMP/IKE
C) AH
D) X.509
Answer: B

Q16. A user connects to a VPN that only tunnels corporate resources and accesses the internet directly. This is an example of:
A) Full tunnel
B) DNS tunneling
C) IP over IP
D) Split tunnel
Answer: D

Q17. Why is **tunnel mode** preferred over **transport mode** for site-to-site VPNs?
A) It is faster
B) It hides internal IP addresses by encrypting the full packet
C) It does not support NAT
D) It compresses packets
Answer: B

Q18. Which protocol is commonly used in **site-to-site VPNs with IPsec** to create tunnels?
A) PPP
B) GRE
C) HTTPS
D) SMTP
Answer: B

Q19. What does IKEv2 provide over IKEv1?
A) No encryption
B) Stateless operation
C) Mobility and multi-homing support
D) Works only on IPv4
Answer: C

Q20. What is a major **risk of split tunneling** if not secured properly?
A) IP duplication
B) DNS leakage
C) Man-in-the-middle attacks from the public network
D) Port blocking
Answer: C

Q21. Which VPN mode is commonly used when **two routers connect two LANs** over the internet?

A) Transport
B) Tunnel
C) GRE-only
D) DMVPN
Answer: B

Q22. Which of these best describes the **Security Association (SA)** in IPsec?
A) Encrypted header
B) Shared key agreement
C) One-way logical connection defining security parameters
D) Transport header
Answer: C

Q23. In IPv6 VPN deployments, **why is NAT not commonly used**?
A) It is insecure
B) IPv6 does not support NAT
C) IPv6 has enough address space for global reach
D) NAT increases IPv6 speed
Answer: C

Q24. Which IPsec encryption algorithm is **recommended for high security**?
A) DES

B) 3DES
C) AES-256
D) RC4
Answer: C

Q25. What is the purpose of NAT-T (NAT Traversal) in IPsec?
A) To change encryption methods
B) To bypass NAT devices by encapsulating ESP in UDP
C) To enable DNS forwarding
D) To hide destination IP
Answer: B

Q26. How does OpenVPN provide **cross-platform support** in hybrid VPNs?
A) Using native SSL modules in each OS
B) Avoids kernel modules entirely
C) Runs as a user-space daemon, independent of OS kernel
D) It doesn’t support cross-platform
Answer: C

Q27. Why might you **combine IPsec and GRE** in a hybrid VPN?
A) To make encryption optional
B) To support multicast routing over an encrypted tunnel

C) To allow FTP over VPN
D) To increase latency
Answer: B

Q28. Which VPN deployment mode is typically considered **most secure** for remote workers using public networks?
A) Split tunnel with NAT
B) Split tunnel with MAC filtering
C) Full tunnel with strong encryption
D) GRE tunnel only
Answer: C

Q29. Which of the following is **false** regarding IPsec tunnel mode?
A) Adds a new IP header
B) Encrypts both IP header and payload
C) Best for inter-LAN connections
D) Less secure than transport mode
Answer: D
Tunnel mode is generally more secure because it hides internal IP addressing.

Q30. What feature must a **firewall** have to properly support IPsec with NAT-T?
A) GRE inspection
B) SSL decryption

C) UDP port 4500 handling
D) ARP support
Answer: C

🔴 Hybrid VPN – Hard-Level MCQs (10 Questions)

Q31. Which cryptographic method does IPsec use in **Perfect Forward Secrecy (PFS)** to ensure key uniqueness per session?
A) AES with static key
B) Pre-shared keys (PSK)
C) Ephemeral Diffie-Hellman
D) RSA key reuse
Answer: C
✅ *Ephemeral Diffie-Hellman generates unique keys per session, ensuring that compromise of one session does not affect others.*

Q32. In a hybrid VPN combining GRE + IPsec, GRE encapsulates IP packets to:
A) Provide encryption
B) Allow multicast and dynamic routing over IPsec
C) Block unauthorized access
D) Increase packet size
Answer: B
✅ *GRE allows protocols like OSPF and BGP to function through IPsec tunnels, which alone don't support multicast.*

Q33. Which of the following correctly explains **split tunneling security risk** in enterprise VPN usage?
A) It causes duplicate packets
B) It bypasses the internal DNS resolution
C) It may allow access from the public internet into the corporate network if the endpoint is compromised
D) It overloads the VPN gateway
Answer: C
 A compromised client can act as a bridge between the internet and internal corporate network.

Q34. In **IPv6 VPN**, why is stateful packet inspection (SPI) more critical compared to IPv4-based VPNs?
A) IPv6 disables all firewalls
B) IPv6 allows multiple NATs
C) All devices can have globally routable addresses
D) IPv6 requires UDP NAT
Answer: C
 Without NAT, IPv6-connected devices are directly accessible unless filtered by SPI or firewalls.

Q35. Which of the following is a correct tunnel configuration for a dual-stack (IPv4+IPv6) hybrid VPN?
A) IPsec over Telnet
B) IPv4 tunnel carrying IPv6 traffic (6in4)
C) NAT64 over IPv6
D) GRE over SSL
Answer: B
 6in4 is a tunneling method that allows IPv6 packets to be encapsulated in IPv4, useful in dual-stack networks.

Q36. How does IKEv2 improve **mobility and multi-homing (MOBIKE)** for VPN clients?
A) It uses static IP addresses
B) It keeps rekeying static
C) It maintains VPN session continuity when client's IP changes (e.g., Wi-Fi to 4G switch)
D) It disables firewall inspection
Answer: C

Q37. In OpenVPN, what does using tls-crypt offer beyond tls-auth?
A) Provides hashing only
B) Obfuscates TLS handshake and encrypts control channel metadata
C) Disables encryption
D) Allows TCP port forwarding
Answer: B
tls-crypt encrypts the entire control channel to prevent metadata leaks and protect against DPI.

Q38. Which combination best describes a **multi-protocol hybrid VPN** for a global enterprise?
A) MPLS + FTP + DNSSEC
B) L2TP + GRE + RIP
C) IPsec for site-to-site + SSL for remote + WireGuard for mobile users
D) UDP + TCP tunneling over ICMP
Answer: C

Q39. A VPN administrator wants to implement **QoS policies over an IPsec tunnel**. Which challenge will they likely face?
A) IPsec encrypts the DSCP field used by QoS
B) IPsec tunnels disable TCP
C) QoS works only with IPv6
D) GRE encapsulation blocks QoS
Answer: A
▯ *IPsec can obscure QoS fields like DSCP unless configured carefully with tunnel mode or GRE support.*

Q40. What is the **primary reason** that full-tunnel VPNs may introduce performance issues in global deployments?
A) They compress the IP headers
B) All user traffic, including public internet access, must pass through the VPN hub
C) They disable routing protocols
D) They bypass DNS resolvers
Answer: B
▯ *Full-tunnel routes all traffic through a central VPN server, adding latency and potential congestion.*